Numerical solution of set of equation

Consider a system of n linear algebraic equations in n unknowns

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2$$
$$\ldots \quad \ldots \quad \ldots \quad \ldots$$
$$a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n = b_n$$

where a_{ij}, $i = 1, 2, \ldots, n$, $j = 1, 2, \ldots, n$, are the known coefficients, b_i, $i = 1, 2, \ldots, n$, are the known right hand side values and x_i, $i = 1, 2, \ldots, n$ are the unknowns to be determined.

In matrix notation we write the system as

$$\mathbf{Ax} = \mathbf{b} \qquad (1)$$

where $$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \ldots & \ldots & \ldots & \ldots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ x_n \end{bmatrix}, \text{ and } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \ldots \\ b_n \end{bmatrix}.$$

The matrix $[\mathbf{A} \mid \mathbf{b}]$, obtained by appending the column $\mathbf{b}$ to the matrix $\mathbf{A}$ is called the *augmented matrix*. That is

$$[\mathbf{A}|\mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ a_{n1} & a_{n2} & \ldots & a_{nn} & b_n \end{array}\right]$$

Example: How to Write a System in Matrix Form

$$x_1 + 10x_2 - x_3 = 3$$
$$2x_1 + 3x_2 + 20x_3 = 7$$
$$10x_1 - x_2 + 2x_3 = 4$$

Solution: We have the matrices as $Ax = B$

$$A = \begin{bmatrix} 1 & 10 & -1 \\ 2 & 3 & 20 \\ 10 & -1 & 2 \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 \\ 7 \\ 4 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 10 & -1 \\ 2 & 3 & 20 \\ 10 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ 4 \end{bmatrix}$$

We have the augmented matrix as

$$\left[\begin{array}{ccc|c} 1 & 10 & -1 & 3 \\ 2 & 3 & 20 & 7 \\ 10 & -1 & 2 & 4 \end{array}\right]$$

We define the following.

(*i*) The system of equations (1) is *consistent* (has at least one solution), if

rank (**A**) = rank [**A** | **b**] = r.

If $r = n$, then the system has unique solution.

If $r < n$, then the system has $(n - r)$ parameter family of infinite number of solutions.

(*ii*) The system of equations (1) is *inconsistent* (has no solution) if

rank (**A**) ≠ rank [**A** | **b**].

We assume that the given system is consistent.

The methods of solution of the linear algebraic system of equations (1) may be classified as direct and iterative methods.

(*a*) *Direct methods* produce the exact solution after a finite number of steps (disregarding the round-off errors). In these methods, we can determine the total number of operations (additions, subtractions, divisions and multiplications). This number is called the *operational count* of the method.

(*b*) *Iterative methods* are based on the idea of successive approximations. We start with an initial approximation to the solution vector $\mathbf{x} = \mathbf{x}_0$, and obtain a sequence of approximate vectors $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k, \ldots$, which in the limit as $k \to \infty$, converge to the exact solution vector $\mathbf{x}$.

Now, we derive some direct methods.

Direct Methods

If the system of equations has some special forms, then the solution is obtained directly. We consider two such special forms.

(*a*) Let $\mathbf{A}$ be a diagonal matrix, $\mathbf{A} = \mathbf{D}$. That is, we consider the system of equations $\mathbf{Dx} = \mathbf{b}$ as

$$\begin{aligned} a_{11}x_1 &= b_1 \\ a_{22}x_2 &= b_2 \\ \ldots \quad \ldots \quad \ldots \quad &\ldots \\ a_{n-1,\,n-1}\,x_{n-1} &= b_{n-1} \\ a_{nn}x_n &= b_n \end{aligned} \tag{2}$$

This system is called a *diagonal system of equations*. Solving directly, we obtain

$$x_i = \frac{b_i}{a_{ii}}, \quad a_{ii} \neq 0, \quad i = 1, 2, \ldots, n. \tag{3}$$

(*b*) Let $\mathbf{A}$ be an upper triangular matrix, $\mathbf{A} = \mathbf{U}$. That is, we consider the system of equations $\mathbf{Ux} = \mathbf{b}$ as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots \quad \ldots \quad & + a_{1n}x_n = b_1 \\ a_{22}x_2 + \ldots \quad \ldots \quad & + a_{2n}x_n = b_2 \\ \ldots \quad \ldots \quad \ldots \quad & \ldots \\ a_{n-1,\,n-1}\,x_{n-1} + a_{n-1,\,n}\,x_n &= b_{n-1} \\ a_{nn}x_n &= b_n \end{aligned} \tag{4}$$

This system is called an *upper triangular system of equations*. Solving for the unknowns in the order $x_n, x_{n-1}, \ldots, x_1$, we get

$$x_n = b_n/a_{nn},$$

$$x_{n-1} = (b_{n-1} - a_{n-1,\,n}x_n)/a_{n-1,\,n-1},$$

$$\ldots \quad \ldots \quad \ldots \quad \ldots$$

$$x_1 = \frac{\left(b_1 - \sum_{j=2}^{n} a_{1,j}x_j\right)}{a_{11}} = \left(b_1 - \sum_{j=2}^{n} a_{1,j}\,x_j\right) \Big/ a_{11} \tag{5}$$

The unknowns are obtained by back substitution and this procedure is called the *back substitution* method.

Therefore, when the given system of equations is one of the above two forms, the solution is obtained directly.

Before we derive some direct methods, we define elementary row operations that can be performed on the rows of a matrix.